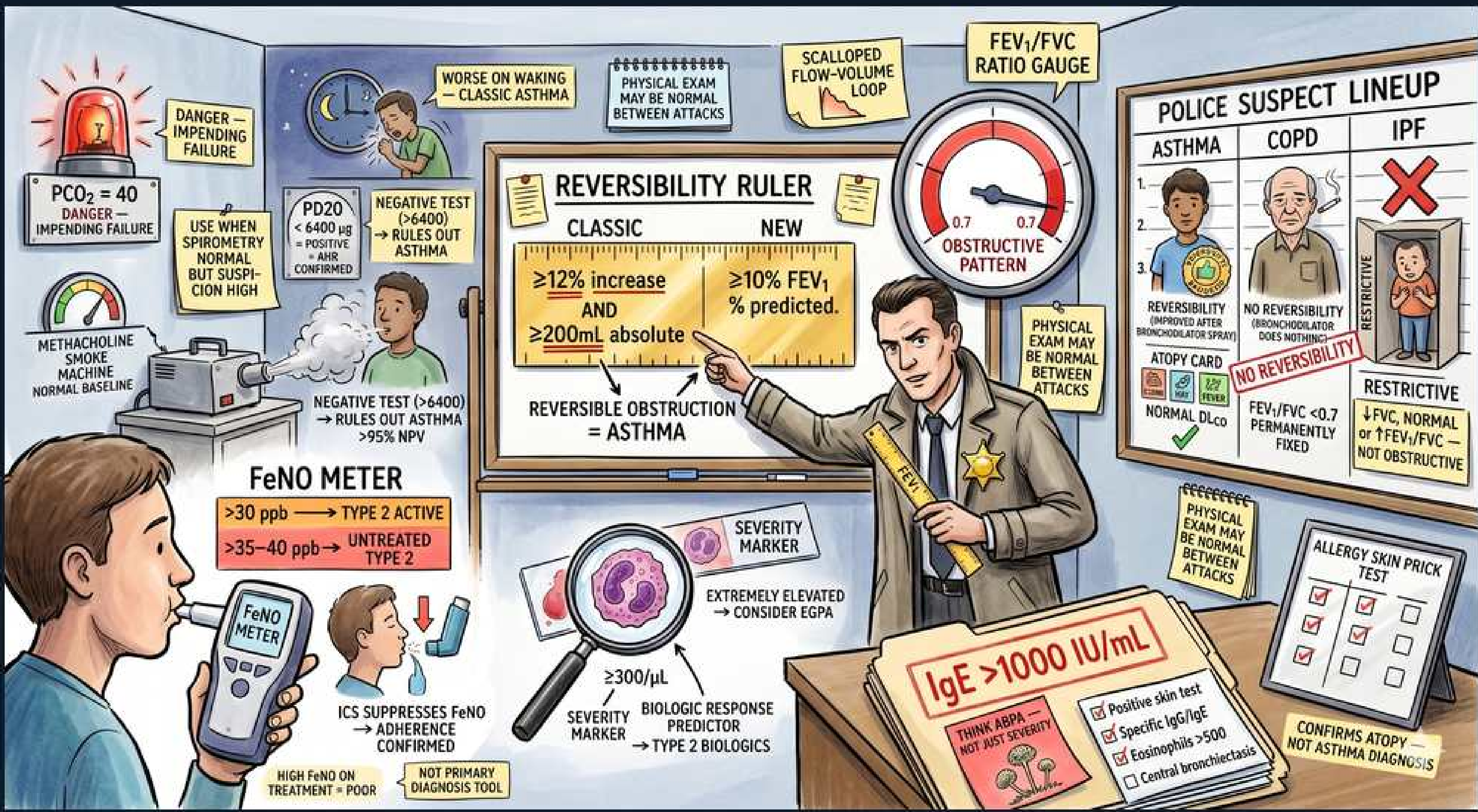
Asthma — Diagnosis & Reversibility Workup



1. Rising PCO2 = impending failure

WHAT YOU SEE

A red alarm light 'PCO2 ↑ → impending failure'.

WHAT IT ENCODES

During a severe attack the tachypneic patient normally has LOW $PaCO_2$ (respiratory alkalosis). A $PaCO_2$ that normalizes (back to ~40) and then rises signals respiratory-muscle fatigue and falling minute ventilation — impending respiratory failure requiring escalation toward intubation.

BOARD TRAP

A 'normal' $PaCO_2$ during a severe attack is ominous, not reassuring — it marks the crossover from hyperventilation to exhaustion.

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2. Worse on waking

WHAT YOU SEE

A figure waking with 'worse on waking — classic asthma'.

WHAT IT ENCODES

Diurnal variability with nocturnal/early-morning symptoms (cough, wheeze, dyspnea waking the patient) reflects circadian dips in airway caliber and is a hallmark of asthma. PEF is characteristically lowest in the morning ('morning dipping'), and >10-20% diurnal PEF variability supports the diagnosis.

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3. Scalloped flow-volume loop

WHAT YOU SEE

A 'scalloped flow-volume loop' diagram.

WHAT IT ENCODES

The expiratory limb of the flow-volume loop is concave/scooped (scalloped) in obstruction — peak flow is reached early then flow collapses as small airways close. This pattern is shared by asthma and COPD; reversibility testing then separates them.

BOARD TRAP

A FLAT/plateaued (fixed) loop on inspiration and expiration suggests a FIXED large-airway obstruction (e.g., tracheal stenosis/tumor), not asthma — shape of the loop localizes the lesion.

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4. FEV1/FVC obstructive gauge

WHAT YOU SEE

A gauge 'FEV1/FVC ratio' pointing to 'obstructive pattern <0.7'.

WHAT IT ENCODES

Spirometry is the cornerstone: a reduced FEV1/FVC ratio (<0.7, or below the lower limit of normal) defines an OBSTRUCTIVE defect. FEV1 falls more than FVC because expiratory airflow is limited. Asthma's obstruction is reversible; COPD's is not.

BOARD TRAP

A NORMAL FEV1/FVC between attacks does NOT exclude asthma — spirometry is frequently normal when the patient is asymptomatic, so a normal study in a suggestive history warrants a methacholine challenge.

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5. Methacholine machine

WHAT YOU SEE

A 'methacholine machine' with 'normal but suspicion high — use when spirometry normal'.

WHAT IT ENCODES

The methacholine bronchoprovocation challenge detects airway HYPERRESPONSIVENESS when baseline spirometry is normal but suspicion is high. A ≥20% fall in FEV1 at a low methacholine dose (PC20) is positive. It is highly SENSITIVE with a high negative predictive value.

BOARD TRAP

A NEGATIVE methacholine test effectively rules OUT asthma (high NPV); a POSITIVE test is less specific (also positive in COPD, allergic rhinitis, recent infection), so it confirms hyperresponsiveness, not necessarily asthma.

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6. Reversibility ruler

WHAT YOU SEE

A ruler 'REVERSIBILITY RULER — classic ≥12% AND ≥200mL; new ≥10% FEV predicted'.

WHAT IT ENCODES

Significant bronchodilator reversibility = post-bronchodilator FEV1 (or FVC) improvement of ≥12% AND ≥200 mL — this confirms reversible airflow obstruction and clinches asthma. (Updated GINA also accepts ≥10% of predicted FEV1.) Both the percent and absolute thresholds must be met under the classic rule.

BOARD TRAP

Reversibility distinguishes asthma from fixed COPD obstruction — but you must satisfy BOTH ≥12% AND ≥200 mL; meeting only one (e.g., 15% but 150 mL) does not count as a positive bronchodilator response.

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7. Police suspect lineup (asthma/COPD/IPF)

WHAT YOU SEE

A 'police suspect lineup': Asthma (reversible), COPD (no reversibility), IPF (restrictive, crossed out).

WHAT IT ENCODES

Differential by PFT pattern: Asthma = obstructive + REVERSIBLE; COPD = obstructive + irreversible (FEV1/FVC <0.7 post-BD); IPF/interstitial disease = RESTRICTIVE (low FVC and TLC, normal/high ratio) with low DLCO. The pattern triages the suspect.

BOARD TRAP

IPF/restrictive disease has a NORMAL or HIGH FEV1/FVC ratio (both FEV1 and FVC fall together) — do not mistake restriction for obstruction; a high ratio rules asthma/COPD out.

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8. FeNO meter

WHAT YOU SEE

A person blowing into a 'FeNO meter' with '>30/35-40 ppb = type 2 active'.

WHAT IT ENCODES

Elevated FeNO (>40-50 ppb high; 25-50 intermediate; IL-13-driven) indicates active eosinophilic/type-2 airway inflammation and predicts corticosteroid responsiveness. It supports the asthma diagnosis and helps select type-2 biologics, but is an adjunct — not a stand-alone diagnostic.

BOARD TRAP

ICS therapy SUPPRESSES FeNO — a low value in a treated patient may reflect good adherence/treatment, not absence of disease; conversely a high FeNO on ICS suggests non-adherence or poor technique.

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9. Eosinophils severity marker

WHAT YOU SEE

A cluster of eosinophils '≥300/μL — biologic response predictor'.

WHAT IT ENCODES

Blood eosinophilia (≥300/μL, or ≥150 at biologic initiation) marks the eosinophilic phenotype, correlates with exacerbation frequency, and predicts response to anti-IL-5 and dupilumab. It guides both prognosis and biologic selection.

BOARD TRAP

Chronic oral steroids and even ICS suppress the eosinophil count — a 'normal' eos in a treated severe asthmatic may understate true type-2 disease; and extremely high eosinophils should prompt thinking of EGPA, not just asthma.

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10. EGPA flag (extremely elevated eos)

WHAT YOU SEE

A note 'extremely elevated → consider EGPA'.

WHAT IT ENCODES

Markedly elevated eosinophils (often >1500/μL or >10%) in an adult-onset asthmatic with systemic features — sinusitis/nasal polyps, mononeuritis multiplex, skin lesions, pulmonary infiltrates, cardiac or renal involvement — suggest EGPA (eosinophilic granulomatosis with polyangiitis / Churg-Strauss), an ANCA-associated (often p-ANCA/MPO) small-vessel vasculitis.

BOARD TRAP

New mononeuritis multiplex, purpura, or migratory infiltrates in an asthmatic point to EGPA, not 'just severe asthma' — and leukotriene-antagonist use has been associated with unmasking EGPA when steroids are tapered.

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11. IgE >1000 / atopy

WHAT YOU SEE

A serum test 'IgE >1000 IU/mL'.

WHAT IT ENCODES

Very high total IgE (often >1000 IU/mL) in an asthmatic with central bronchiectasis, recurrent infiltrates, and Aspergillus sensitization raises concern for ABPA (allergic bronchopulmonary aspergillosis), treated with systemic steroids ± antifungals. Moderately elevated IgE simply supports atopy and guides omalizumab dosing.

BOARD TRAP

IgE confirms atopy and helps dose omalizumab but is NOT a diagnostic test for asthma itself — and a markedly high IgE with worsening asthma + bronchiectasis should trigger an ABPA workup (Aspergillus-specific IgE/IgG, eosinophilia).

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12. Allergy skin-prick test

WHAT YOU SEE

An 'allergy skin prick test' confirming atopy, noted 'not asthma diagnostic'.

WHAT IT ENCODES

Allergy skin-prick (or specific-IgE) testing identifies sensitizing aeroallergens to guide trigger avoidance, allergen immunotherapy, and biologic choice (e.g., omalizumab requires perennial allergen sensitization). It characterizes the allergic phenotype, not the diagnosis.

BOARD TRAP

A positive skin-prick confirms ATOPY/sensitization, not asthma — many sensitized people never develop asthma, and non-allergic (eosinophilic) asthma exists with negative skin tests.
